

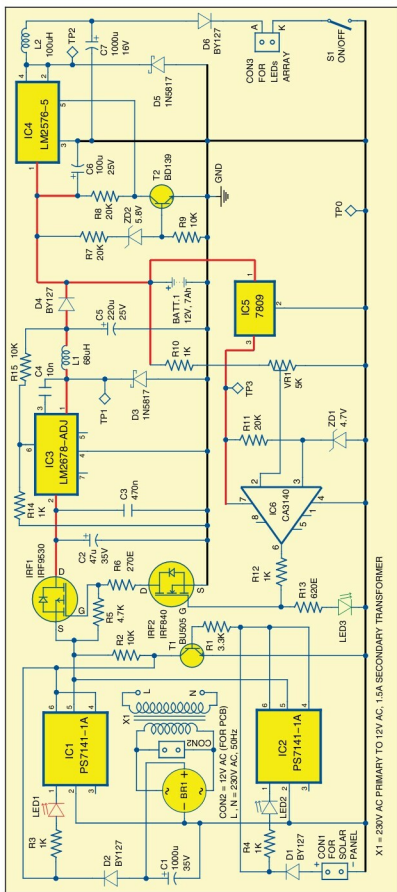


Fig. 1: Circuit diagram of the solar emergency lamp

Solar Emergency Lamp



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Solar-powered lights are a popular alternative to electric-powered lights. The solar emergency lamp described here has 60 white LEDs. It is powered either by solar power or AC mains or 12V, SMF battery; under normal operation, the lamp is powered by the battery.

When the battery voltage drops to 10.5V, the battery-charge-controller circuit goes into operation. Selection between a solar panel and AC mains for charging is done with the help of solid-state relays. If both sources are connected, the system selects solar power over AC mains for charging the battery.

Circuit and working

Fig. 1 shows the circuit diagram of the solar emergency lamp. It is built around a solar panel connected across connector CON1, 15V AC connected across connector CON2, 12V, 7Ah battery, solid-state relay PS7141-1A (IC1 and IC2), p-channel MOSFET IRF9530 (IRF1) and n-channel MOSFET IRF840 (IRF2), IC LM2678-ADJ switching regulator (IC3), 5V switching regulator IC LM2576 (IC4), 9V voltage series regulator 7809 (IC5), op-amp comparator IC CA3140 (IC6), a lamp formed by an array of 60 white LEDs and a few other components. The circuit is divided into two parts: a solar charger and an LED array.

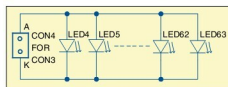


Fig. 2: Circuit diagram of the LED array